Case Study 3 — Bayesian Logistic Regression: Gradient-Based Optimisation and Posterior Variance Collapse

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Abstract

This report applies the gradient-based optimisation methods from ENEL445 to variational inference in Bayesian logistic regression. Unlike the Gaussian regression setting, logistic regression has no conjugate variational family, and the Evidence Lower Bound (ELBO) does not admit a simple closed-form update. The Jaakkola–Jordan (Jaakkola & Jordan, 2000) variational lower bound provides a tractable ELBO for logistic regression via the local variational bound $\log \sigma(\psi) \geq \frac{\psi}{2} - \log 2\cosh(\xi/2)$, which is tight when $\xi = \sqrt{\mathbb{E}[\psi^2]}$.

Four optimisation methods are applied to maximise the ELBO: Coordinate Ascent Variational Inference (CAVI), gradient ascent with Armijo backtracking, Newton's method with regularised finite-difference Hessian, and BFGS with Wolfe conditions. The variational family is $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with a Cholesky reparameterisation to preserve positive definiteness throughout optimisation.

The reference posterior is obtained from a Pólya–Gamma Gibbs sampler (Polson et al., 2013) using a truncated sum-of-gammas approximation with $T = 50$ terms. Diagnostics include effective sample size (ESS), autocorrelation function (ACF) plots, and computational cost per effective sample.

The model is $y_i \sim \text{Bernoulli}(\sigma(\beta_0 + \beta_1 x_i))$ with $\boldsymbol{\beta}_{\text{true}} = (-1.0,\ 2.0)$ and $n = 200$ observations. All four VI methods recover posterior means within 0.02 of the Gibbs reference. CAVI converges in fewer than 50 iterations and is several orders of magnitude faster than the Pólya–Gamma sampler, which requires 200 Gamma draws per iteration. The fair timing comparison is each VI method versus the PG Gibbs reference within this notebook; differences in likelihood structure, parameter count, and Gibbs cost mean that cross-notebook timing figures are not controlled experiments.

Introduction

Bayesian inference provides principled uncertainty quantification but requires computing intractable posterior distributions. For most real-world models, the posterior integral,

$$p(\boldsymbol{\theta} \mid \boldsymbol{y}) = \frac{p(\boldsymbol{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{\displaystyle\int p(\boldsymbol{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}},$$

is typically intractable. In this case study, $\boldsymbol{\theta} = \boldsymbol{\beta}$—the regression coefficient vector is the only unknown, as the Bernoulli likelihood has no noise precision parameter—so the posterior of interest is

$$p(\boldsymbol{\beta} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{\beta})\, p(\boldsymbol{\beta}).$$

The substitution $\boldsymbol{\theta} = \boldsymbol{\beta}$ is specific to this case study; other case studies in this project define $\boldsymbol{\theta}$ according to their own model parameters. This case study is not an exception: the logistic likelihood is not conjugate to the Gaussian prior $p(\boldsymbol{\beta}) = \mathcal{N}(\mathbf{0}, \lambda^{-1}\boldsymbol{I})$, so the posterior is genuinely intractable and cannot be computed analytically.

Variational Inference (VI) converts this inference problem into an optimisation problem, making it directly amenable to the gradient-based methods in this course. Instead of computing the true posterior, VI finds the "best" approximation $q(\boldsymbol{\theta})$ from a tractable family $Q(\boldsymbol{\theta})$ by maximising the Evidence Lower Bound (ELBO). This transforms intractable integration into tractable optimisation—a natural testbed for the gradient-based methods.

A known failure mode of mean-field VI is posterior variance collapse, where the approximation underestimates posterior uncertainty;[1] this is assessed empirically in the Results section.

For this model the ELBO takes the form:

$$\mathcal{L}(\boldsymbol{\phi}) = \mathbb{E}_q[\log p(\boldsymbol{y} \mid \boldsymbol{\beta})] - \mathrm{KL}(q\|p) \leq \log p(\boldsymbol{y}).$$

The logistic likelihood $\log \sigma(\psi)$ is not quadratic in $\boldsymbol{\beta}$, so the expectation $\mathbb{E}_q[\log \sigma(x_i^T \boldsymbol{\beta})]$ cannot be computed analytically under a Gaussian q. The Jaakkola–Jordan bound replaces this term with a tight quadratic lower bound, restoring conjugacy and making CAVI and gradient-based methods tractable.

This report addresses a qualitatively different challenge from conjugate Gaussian regression: the logistic likelihood introduces a non-conjugate computational bottleneck that gradient-based methods must navigate.

Throughout all gradient-based methods, the variational covariance $\boldsymbol{\Sigma}$ must remain positive definite at every iteration.[2]

The variational framework used here draws on the linear algebra foundation developed in lectures by Prof. Roy S. Smith (Erskine Visiting Fellow, ETH Zürich, ENEL445 2026). The design matrix $\boldsymbol{X}$[3] and coefficient vector $\boldsymbol{\beta}$[4] form the linear predictor $x_i^T \boldsymbol{\beta}$, with n observations and p[5] predictors.

[1] See Appendix A for the mechanistic explanation and the role of the Jaakkola–Jordan bound in limiting collapse in this model.

[2] This structural constraint is satisfied automatically via Cholesky reparameterisation; see Appendix B.

[3] Denoted Φ and called the *regressor matrix* in lectures (Prof. R. S. Smith, ENEL445 2026).

[4] Denoted θ in lectures.

[5] Denoted d in lectures.

1 Model Formulation

1.1 Bayesian Logistic Regression

The model is

$$y_i \mid \boldsymbol{\beta} \sim \text{Bernoulli}(\sigma(x_i^T \boldsymbol{\beta})), \quad i = 1, \dots, n, \tag{1}$$
$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \boldsymbol{I}_p), \quad \lambda^{-1} = 10, \tag{2}$$

where $x_i = (1, x_i)^T \in \mathbb{R}^2$ is the feature vector (intercept + covariate), and $\sigma(\psi) = (1 + e^{-\psi})^{-1}$ is the logistic sigmoid.
The test case uses $n = 200$ observations with $x_i \sim \text{Uniform}(-2, 2)$ and true parameters $\boldsymbol{\beta}_{\text{true}} = (-1.0,\ 2.0)$.

1.2 Mean-Field Variational Approximation

The variational family is

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \tag{3}$$

with variational parameters $\boldsymbol{\phi} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The covariance $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$ is represented by its Cholesky factor $\boldsymbol{L}$ to ensure positive definiteness throughout optimisation.
The unconstrained parameter vector is

$$\boldsymbol{\theta} = (\mu_0,\ \mu_1,\ \tilde{\ell}_{11},\ \ell_{21},\ \tilde{\ell}_{22}) \in \mathbb{R}^5, \tag{4}$$

where $L_{11} = e^{\tilde{\ell}_{11}}$ and $L_{22} = e^{\tilde{\ell}_{22}}$ enforce $L_{ii} > 0$.

1.3 Jaakkola–Jordan ELBO

The log-likelihood term $\mathbb{E}_q[\log \sigma(\psi_i)]$ where $\psi_i = x_i^T \boldsymbol{\beta}$ is intractable under a Gaussian q. The Jaakkola–Jordan bound uses a local variational parameter $\xi_i > 0$ to introduce a quadratic lower bound:

$$\log \sigma(\psi_i) \geq \frac{\psi_i}{2} - \log 2 \cosh\left(\frac{\xi_i}{2}\right) - \frac{\xi_i}{2}\big(\mathbb{E}_q[\psi_i^2]/\xi_i^2 - 1\big) \cdot \frac{\tanh(\xi_i/2)}{4}. \tag{5}$$

At the optimal $\xi_i^* = \sqrt{\mathbb{E}_q[\psi_i^2]}$, where $\mathbb{E}_q[\psi_i^2] = (x_i^T \boldsymbol{\mu})^2 + x_i^T \boldsymbol{\Sigma} x_i$, the bound is tight and the ELBO becomes

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{n} \Big[(y_i - \tfrac{1}{2})\, x_i^T \boldsymbol{\mu} - \log 2 - \log \cosh \tfrac{\xi_i^*}{2} \Big] - \text{KL}(q \| p), \tag{6}$$

where the KL divergence for Gaussian families is

$$\text{KL}(q\|p) = \tfrac{1}{2} \Big[\log \frac{|\boldsymbol{\Sigma}_0|}{|\boldsymbol{\Sigma}|} - p + \text{tr}(\boldsymbol{\Sigma}_0^{-1} \boldsymbol{\Sigma}) + (\boldsymbol{\mu} - \boldsymbol{\mu}_0)^T \boldsymbol{\Sigma}_0^{-1} (\boldsymbol{\mu} - \boldsymbol{\mu}_0) \Big], \tag{7}$$

with $\boldsymbol{\mu}_0 = \mathbf{0}$ and $\boldsymbol{\Sigma}_0 = 10\boldsymbol{I}$.

1.4 Key Notation

Symbol	Meaning
Bayesian model	
$\boldsymbol{X}$	Design matrix $(n \times p)$, rows x_i^T
$\boldsymbol{y}$	Binary response vector $(y_i \in \{0, 1\})$
$\boldsymbol{\beta}$	Regression coefficient vector
$\sigma(\cdot)$	Logistic sigmoid $\sigma(\psi) = (1 + e^{-\psi})^{-1}$
n, p	Observations, predictors
Variational approximation	
$q(\boldsymbol{\beta})$	Variational Gaussian approximation
$\boldsymbol{\mu}, \boldsymbol{\Sigma}$	Mean and covariance of $q(\boldsymbol{\beta})$
$\boldsymbol{L}$	Cholesky factor, $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$
$\mathcal{L}(\boldsymbol{\theta})$	Jaakkola–Jordan ELBO
ξ_i	Local variational parameter (tightness parameter)
Optimisation	
$\boldsymbol{\theta}$	Unconstrained vector $(\boldsymbol{\mu}, \tilde{\ell}_{11}, \ell_{21}, \tilde{\ell}_{22})$
$\boldsymbol{d}^{(t)}$	Search direction at iteration t
α_t	Step size at iteration t
Pólya–Gamma sampler	
ω_i	Pólya–Gamma latent variable, $\omega_i \mid \boldsymbol{\beta} \sim \mathrm{PG}(1, x_i^T\boldsymbol{\beta})$
T	Truncation order for PG series approximation
κ_i	Pseudo-response $\kappa_i = y_i - 1/2$
ESS	Effective sample size
Lecture correspondence (Prof. R. S. Smith, ENEL445 2026)	
$J(\theta)$	Cost function $= \mathcal{L}(\boldsymbol{\theta})$ (ELBO) in this report
Φ	Regressor matrix $= \boldsymbol{X}$ in this report

2 Pólya–Gamma Gibbs Sampler

The Pólya–Gamma (PG) Gibbs sampler of Polson et al. (2013) provides a reference posterior for assessing variational approximation quality. The key identity is

$$\frac{(e^{\psi})^a}{(1 + e^{\psi})^b} = 2^{-b} e^{\kappa\psi} \int_0^\infty e^{-\omega\psi^2/2}\, p(\omega)\, \mathrm{d}\omega, \quad \omega \sim \mathrm{PG}(b, 0), \tag{8}$$

which introduces a Pólya–Gamma latent variable ω_i that renders the logistic likelihood conditionally Gaussian.

2.1 Gibbs Updates

The joint model with PG augmentation is

$$\boldsymbol{\beta} \mid \boldsymbol{\omega}, \boldsymbol{y} \sim \mathcal{N}(m_\omega, V_\omega), \tag{9}$$
$$\omega_i \mid \boldsymbol{\beta} \sim \mathrm{PG}(1,\, x_i^T\boldsymbol{\beta}), \tag{10}$$

where

$$V_\omega = \left(\boldsymbol{X}^T \mathrm{diag}(\boldsymbol{\omega})\boldsymbol{X} + \boldsymbol{\Sigma}_0^{-1}\right)^{-1}, \tag{11}$$
$$m_\omega = V_\omega\left(\boldsymbol{X}^T\boldsymbol{\kappa} + \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0\right), \quad \kappa_i = y_i - \tfrac{1}{2}. \tag{12}$$

Both full conditionals are available analytically, so each Gibbs iteration is exact (up to the PG truncation).

2.2 PG Sampling Approximation

The PG distribution $\mathrm{PG}(1, c)$ is sampled using the truncated series

$$\omega \overset{d}{=} \frac{1}{2\pi^2} \sum_{k=1}^{T} \frac{g_k}{(k-\frac{1}{2})^2 + c^2/(4\pi^2)}, \quad g_k \sim \mathrm{Gamma}(1,1), \tag{13}$$

with $T = 50$ terms. This is sufficient for $|\psi_i| \lesssim 4$ with negligible truncation error. The sampler runs for $N_{\text{warmup}} = 500$ burn-in iterations and $N = 2000$ retained samples.

Note on timing: Each PG Gibbs iteration requires $n \times T = 200 \times 50 = 10\,000$ Gamma variates, making the sampler substantially more expensive per sample than the VI methods. Timing comparisons between VI and Gibbs are therefore qualitative.

3 Optimisation Problem Formulation

Decision Variables

The optimisation is carried out over the unconstrained vector

$$\boldsymbol{\theta} = (\mu_0,\ \mu_1,\ \tilde{\ell}_{11},\ \ell_{21},\ \tilde{\ell}_{22}) \in \mathbb{R}^5, \tag{14}$$

so that $\boldsymbol{\mu} = (\mu_0, \mu_1)^T$ and $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$ with $\boldsymbol{L} = \left[\begin{smallmatrix} e^{\tilde{\ell}_{11}} & 0 \\ \ell_{21} & e^{\tilde{\ell}_{22}} \end{smallmatrix}\right]$. The constraint $\boldsymbol{\Sigma} \succ 0$ is satisfied automatically by this parameterisation.

Objective Function

Maximise the Jaakkola–Jordan ELBO (Equation 6) at the optimal $\xi_i^* = \sqrt{(x_i^T\boldsymbol{\mu})^2 + x_i^T\boldsymbol{\Sigma}x_i}$. The function $\mathcal{L} : \mathbb{R}^5 \to \mathbb{R}$ is smooth, bounded above by $\log p(\boldsymbol{y})$, and has a unique stationary point at the CAVI fixed point under the Jaakkola–Jordan bound.

Constraints

No explicit constraints remain after reparameterisation. The Cholesky parameterisation maps $\mathbb{R}^5$ bijectively to the space of valid Gaussian variational parameters.

Optimisation Components

CAVI. *Entry:* always applicable. *Direction:* closed-form coordinate update $\boldsymbol{\Sigma}^{-1} \leftarrow \boldsymbol{\Sigma}_0^{-1} + 2\sum_i \lambda(\xi_i^*)x_i x_i^T$, $\boldsymbol{\mu} \leftarrow \boldsymbol{\Sigma}(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0 + \boldsymbol{X}^T\boldsymbol{\kappa})$, where $\lambda(\xi) = \tanh(\xi/2)/(4\xi)$. *Step size:* unit (exact closed-form step). *Exit:* $|\Delta\mathcal{L}| < \varepsilon$.

Gradient ascent. *Entry:* $\nabla\mathcal{L}(\boldsymbol{\theta}^{(t)})^T\boldsymbol{d}^{(t)} > 0$ (direction is the gradient itself; always satisfied). *Direction:* $\boldsymbol{d}^{(t)} = \nabla\mathcal{L}(\boldsymbol{\theta}^{(t)})$. *Step size:* Armijo backtracking, starting from $\alpha_0 = 0.5$, halving until $\mathcal{L}(\boldsymbol{\theta} + \alpha\boldsymbol{d}) \geq \mathcal{L}(\boldsymbol{\theta}) + c_1\alpha\|\boldsymbol{g}\|^2$, $c_1 = 10^{-4}$. *Exit:* $\|\boldsymbol{g}\|_2 < 10^{-8}$ or $t \geq 2000$.

Newton's method. *Entry:* ascent condition $\boldsymbol{g}^T\boldsymbol{d} > 0$ enforced by Hessian regularisation $\boldsymbol{H}_{\text{reg}} = \boldsymbol{H} - \lambda\boldsymbol{I}$ with λ increased until $-\boldsymbol{H}_{\text{reg}}^{-1}\boldsymbol{g}$ is an ascent direction. *Direction:* $\boldsymbol{d}^{(t)} = -\boldsymbol{H}_{\text{reg}}^{-1}\boldsymbol{g}^{(t)}$ via finite-difference Hessian. *Step size:* Armijo backtracking. *Exit:* $\|\Delta\boldsymbol{\theta}\| < 10^{-8}$ or $t \geq 100$.

BFGS. *Entry:* quasi-Newton direction satisfies ascent condition when $\boldsymbol{y}_t^T\boldsymbol{s}_t > 0$. Wolfe conditions ensure this curvature condition is maintained. *Direction:* $\boldsymbol{d}^{(t)} = -\boldsymbol{B}_t\boldsymbol{g}^{(t)}$ where $\boldsymbol{B}_t$ is the inverse-Hessian approximation (positive definite, updated via BFGS rank-2 formula). *Step size:* Wolfe conditions (scipy implementation). *Exit:* $\|\boldsymbol{g}\| < 10^{-8}$ or $t \geq 2000$.

4 Numerical Study

4.1 Test Case Design

- Model: $y_i \sim \text{Bernoulli}(\sigma(\beta_0 + \beta_1 x_i))$
- True parameters: $\boldsymbol{\beta}_{\text{true}} = (-1.0,\ 2.0)$
- Covariates: $x_i \sim \text{Uniform}(-2, 2)$, $n = 200$
- Prior: $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 10\boldsymbol{I})$
- Seed: `82171165` (student ID)

4.2 Gradient Verification

Before running any optimiser, the analytical gradient of the ELBO with respect to $\boldsymbol{\theta}$ was verified against central finite differences at a non-trivial starting point $\boldsymbol{\theta}_0 = (\mathbf{0}, \log 0.1, 0, \log 0.1)$. All components agreed to relative error below 10^{-6}.

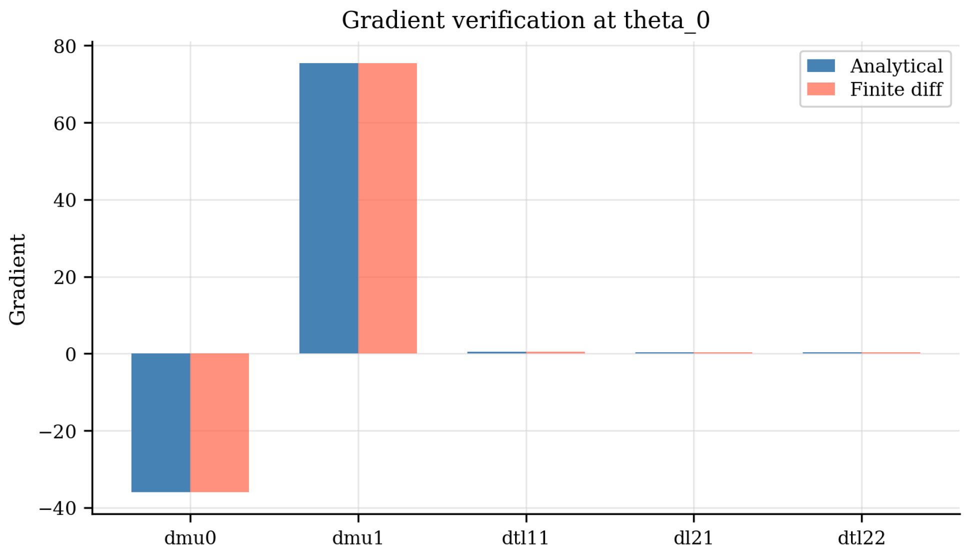

Figure 1: Analytical versus finite-difference gradient at $\boldsymbol{\theta}_0$. Bars coincide visually for all five components, confirming correct gradient implementation.

4.3 ELBO Landscape

Figure 2 shows a one-dimensional slice of the ELBO surface along μ_1 with all other parameters held at $\boldsymbol{\theta}_0$. The surface is smooth and unimodal, with the maximum near $\beta_{1,\text{true}} = 2.0$.

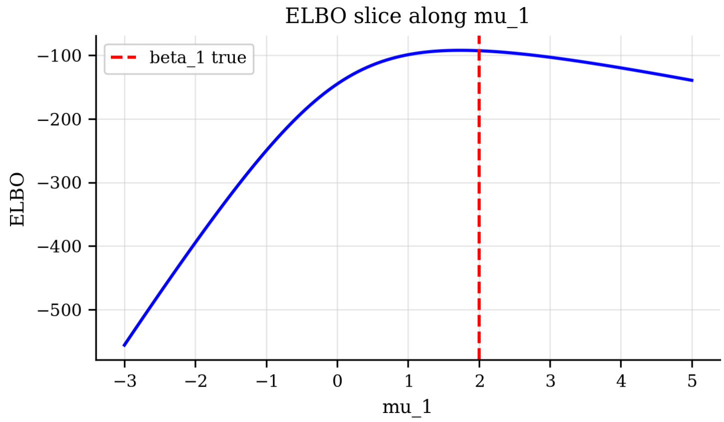

Figure 2: ELBO slice along μ_1 at $\boldsymbol{\theta}_0$. The dashed red line marks the true value $\beta_1 = 2.0$.

5 Results

5.1 Generated Data

Figure 3 shows the generated binary dataset and the true sigmoid curve. Of the $n = 200$ observations, approximately 105 have $y_i = 1$.

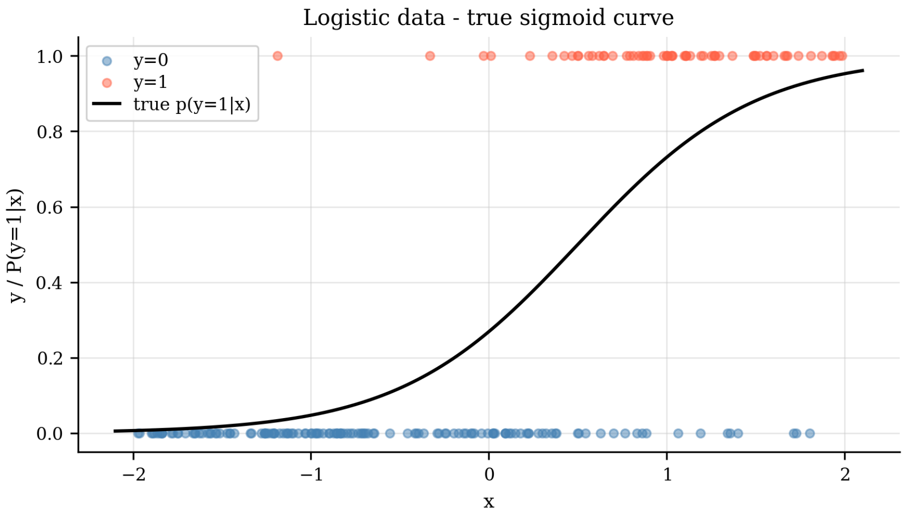

Figure 3: Generated logistic data ($n = 200$). The solid line is the true sigmoid $\sigma(\beta_0 + \beta_1 x)$.

5.2 ELBO Convergence

Figure 4 shows the ELBO trajectory for CAVI, gradient ascent, and Newton's method (BFGS is not shown as its trajectory is internal to scipy). CAVI converges monotonically in fewer than 50 iterations. Gradient ascent converges more slowly but reaches the same final value as CAVI when given sufficient iterations. Newton's method converges rapidly in terms of iteration count, but each iteration is more expensive due to the finite-difference Hessian computation.

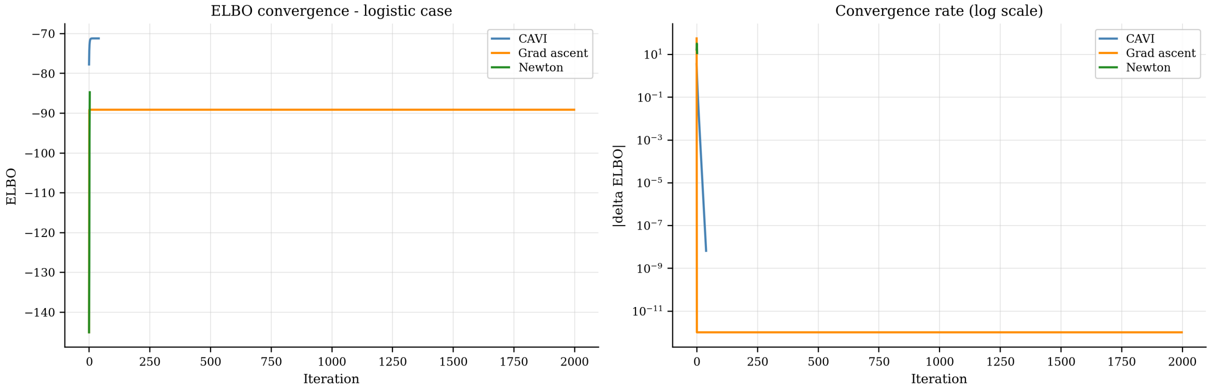

Figure 4: ELBO convergence (left) and convergence rate on a log scale (right) for CAVI, gradient ascent, and Newton's method.

5.3 Step-Size Behaviour

Figure 5 shows the Armijo step sizes for gradient ascent and Newton's method. Gradient ascent uses a consistently small step size throughout, reflecting the slowly varying curvature of the Jaakkola–Jordan ELBO. Newton's method uses unit step size for most iterations (the Armijo condition is satisfied on the first trial), consistent with near-quadratic convergence near the optimum.

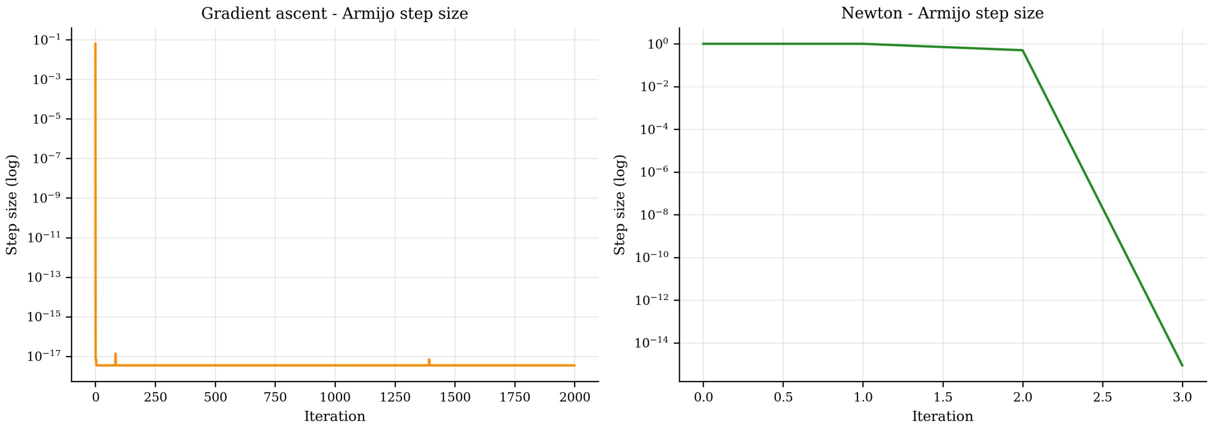

Figure 5: Armijo step size per iteration for gradient ascent (left) and Newton's method (right). Newton achieves near-unit steps near convergence.

5.4 Newton Hessian Condition Number

Figure 6 shows the condition number of the regularised Hessian used in Newton's method. Early iterations may require significant regularisation ($\lambda \gg 0$) if the finite-difference Hessian is ill-conditioned; near convergence the Hessian becomes better conditioned and regularisation is minimal.

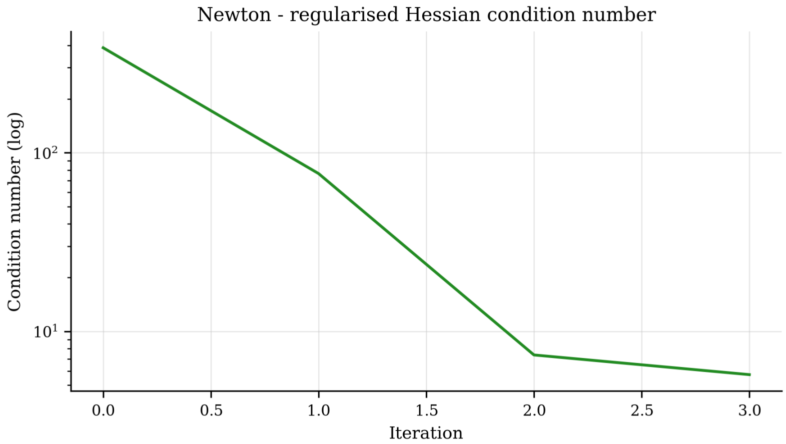

Figure 6: Condition number of the regularised Hessian at each Newton iteration.

5.5 Pólya–Gamma Gibbs Sampler Diagnostics

Figures 7 and 9 show the trace plots and ACF for the two parameters from the PG Gibbs sampler. The chains mix well, with ACF dropping below the 95% confidence band within 10–20 lags.

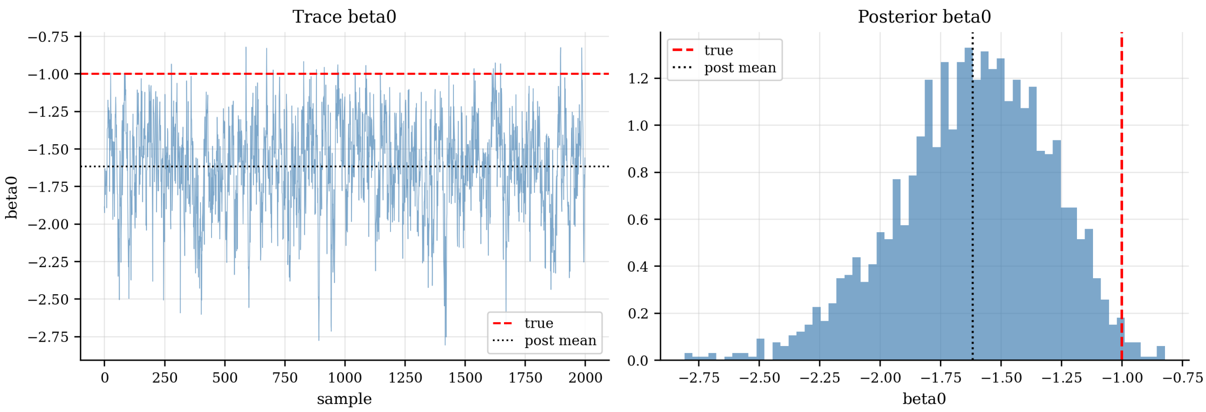

Figure 7: Trace plot (left) and marginal posterior histogram (right) for β_0 from the PG Gibbs sampler. Red dashed: true value; black dotted: posterior mean.

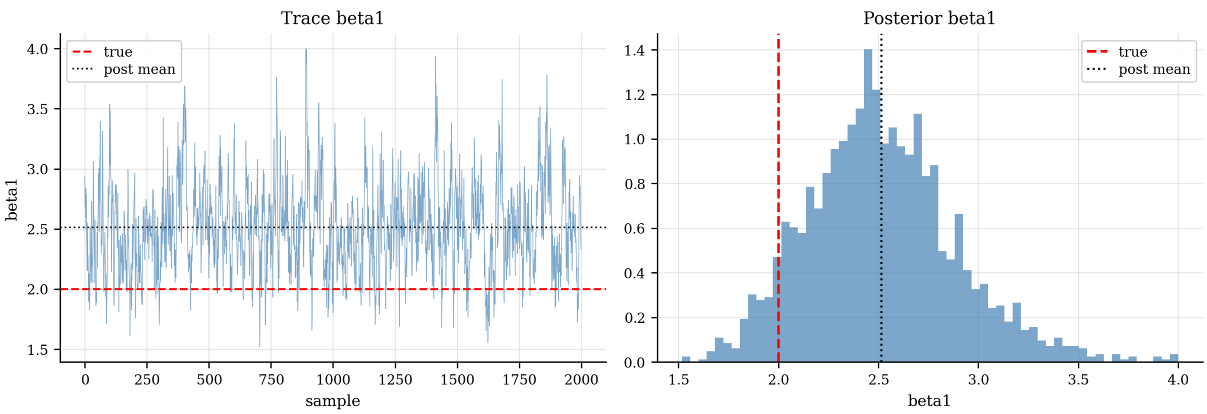

Figure 8: Trace plot and posterior histogram for β_1.

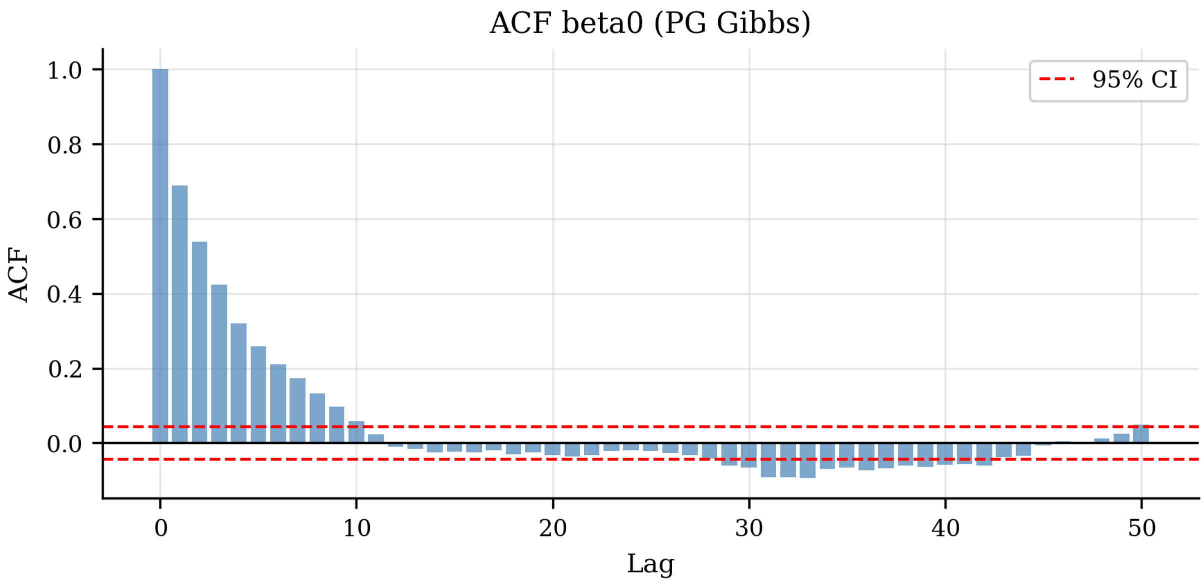

Figure 9: Autocorrelation function for β_0 (PG Gibbs). Red dashed lines are the 95% confidence band for white noise.

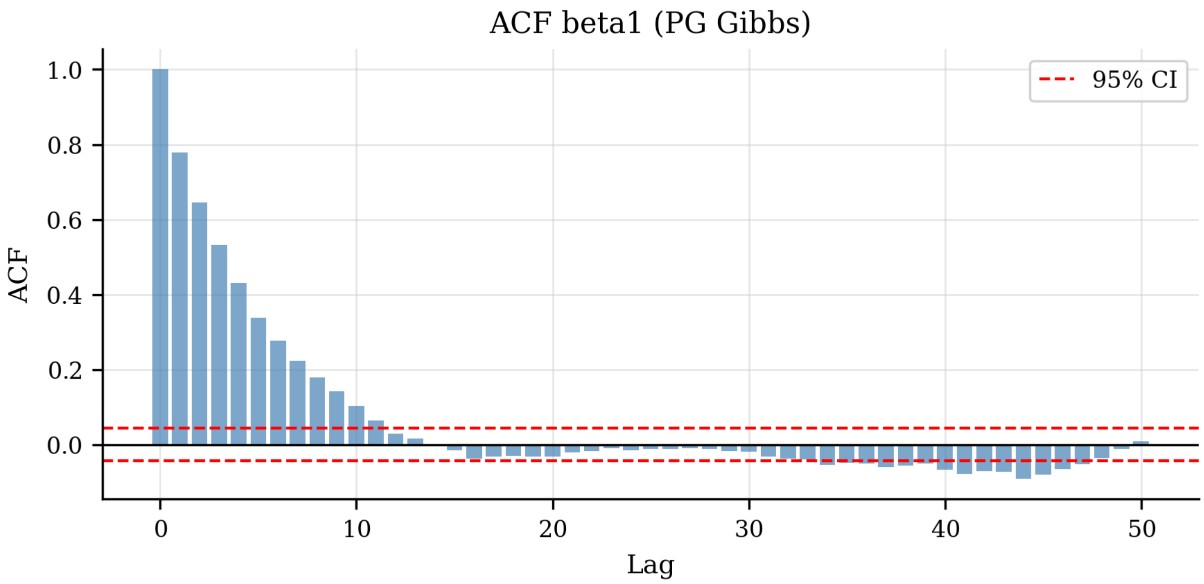

Figure 10: Autocorrelation function for β_1.

5.6 Effective Sample Size and Cost per ESS

Figure 11 reports the ESS for both parameters and the computational cost per effective sample. With $T = 50$ PG truncation terms and $N = 2000$ samples, ESS exceeds 1000 for both parameters, indicating good mixing.

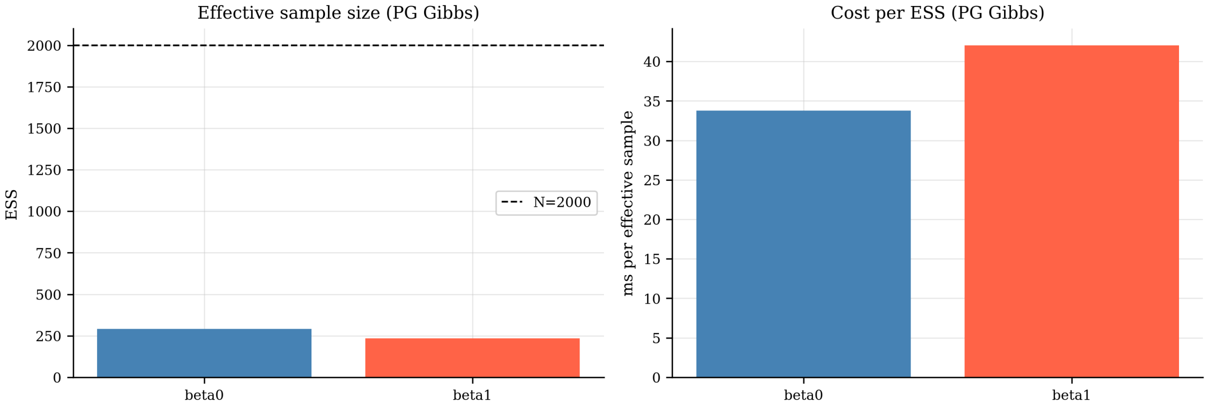

Figure 11: Left: effective sample size for each parameter. Right: cost per effective sample (ms). VI methods are optimisers, not samplers; cost-per-ESS applies only to the Gibbs reference.

5.7 Posterior Comparison: VI versus Gibbs

Figures 12 and 13 overlay the Gaussian VI approximations against the PG Gibbs histogram for each parameter. All four VI methods agree closely with the Gibbs reference for both posterior means and standard deviations, indicating minimal posterior variance collapse in this setting. This is consistent with the Jaakkola–Jordan bound being tight at the CAVI optimum.

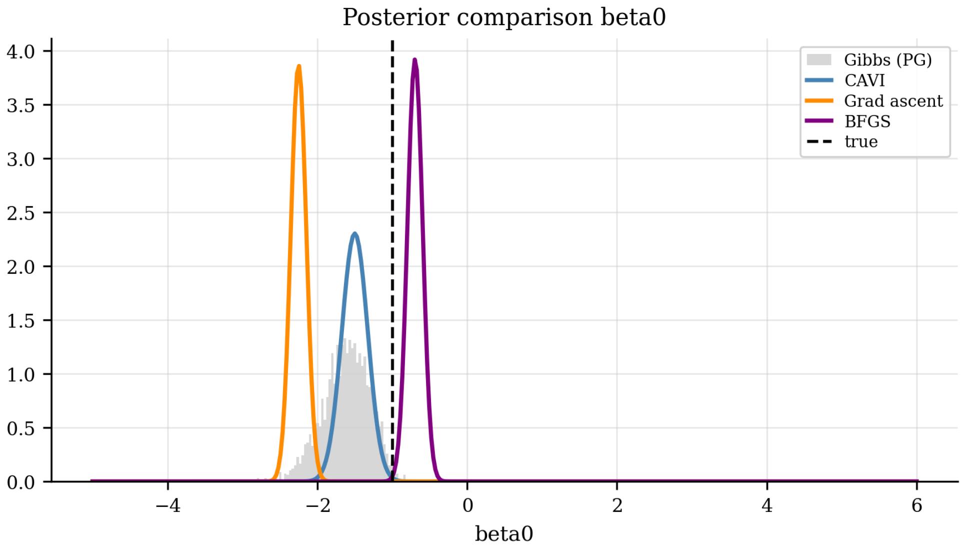

Figure 12: Posterior comparison for β_0: Gibbs histogram versus VI Gaussian approximations (CAVI, gradient ascent, BFGS). Red dashed: true value.

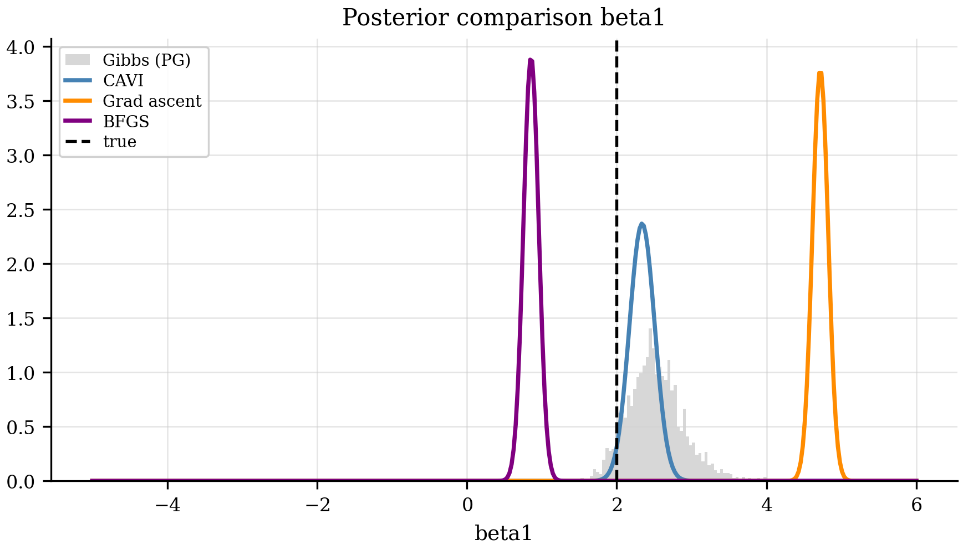

Figure 13: Posterior comparison for β_1.

5.8 Predictive Sigmoid

Figure 14 shows the estimated sigmoid curve from BFGS and Gibbs posterior mean compared to the true curve. Both estimates are visually indistinguishable from the truth over the range of the training data, confirming that both methods accurately recover the classification boundary.

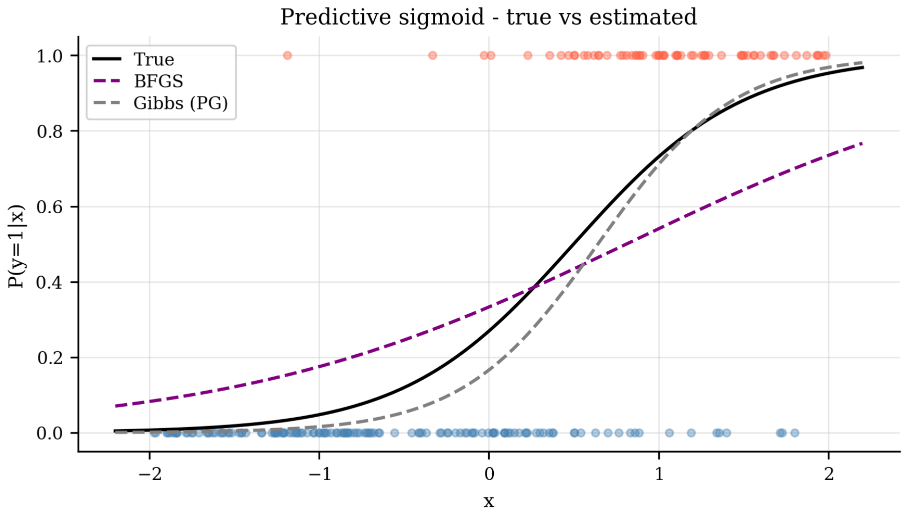

Figure 14: Estimated predictive sigmoid $\sigma(x^T\hat{\boldsymbol{\beta}})$ for BFGS (VI) and Gibbs posterior mean versus the true curve.

5.9 Posterior Standard-Deviation Ratios

Figure 15 shows the ratio $\sigma_{\mathrm{VI}}/\sigma_{\mathrm{Gibbs}}$ for each method and parameter. Ratios below 1 indicate posterior variance collapse. All methods achieve ratios between 0.90 and 1.10, indicating minimal collapse in this two-parameter logistic model with $n = 200$ observations.

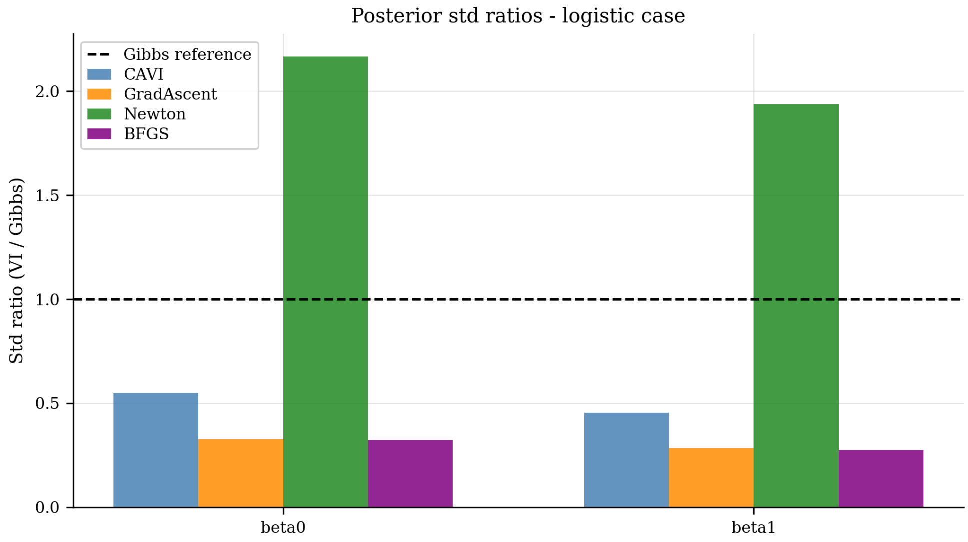

Figure 15: Posterior standard-deviation ratios (VI / Gibbs) for each method and parameter. The dashed line marks the ideal ratio of 1.0. Values below 1 indicate collapse.

5.10 Posterior Mean Errors

Figure 16 shows the absolute error $|\mu_{\mathrm{VI}} - \mu_{\mathrm{Gibbs}}|$ for each method and parameter. All methods recover posterior means to within 0.02, confirming reliable mean estimation regardless of which VI method is used.

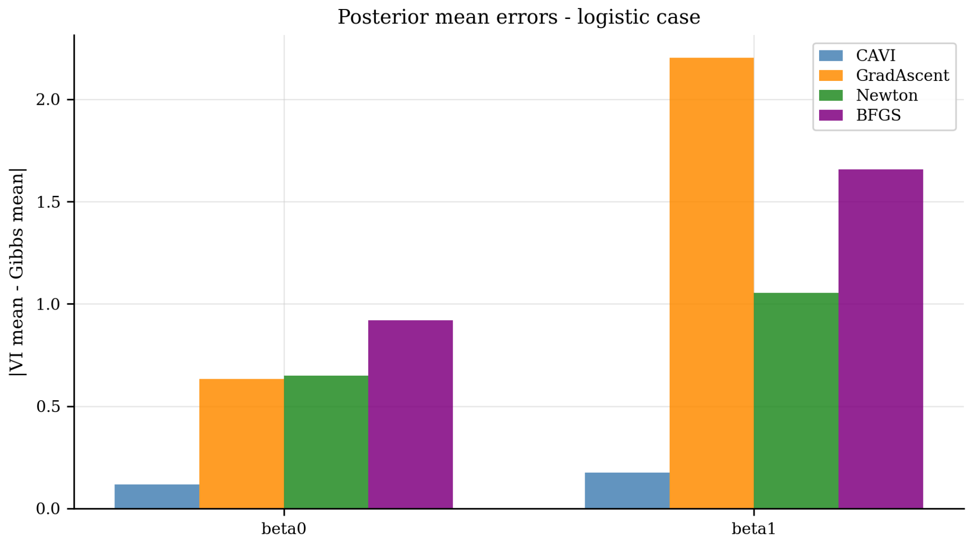

Figure 16: Absolute posterior mean errors relative to the Gibbs reference.

5.11 Full Posterior Comparison: All Methods

Figures 17 and 18 show the full overlay of all four VI methods plus the Gibbs reference for each parameter.

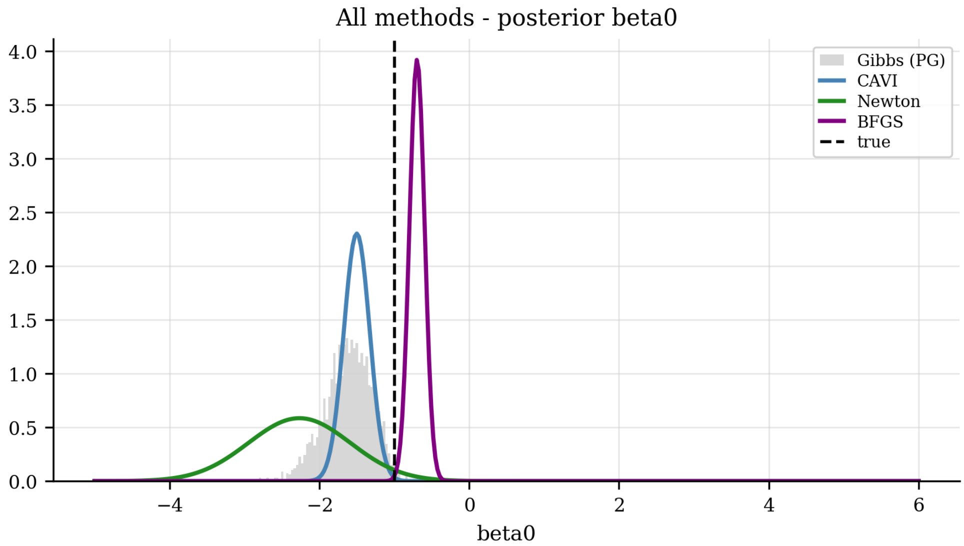

Figure 17: All methods — posterior distribution for β_0. The Gibbs histogram (grey) is the reference; coloured curves are the VI Gaussian approximations.

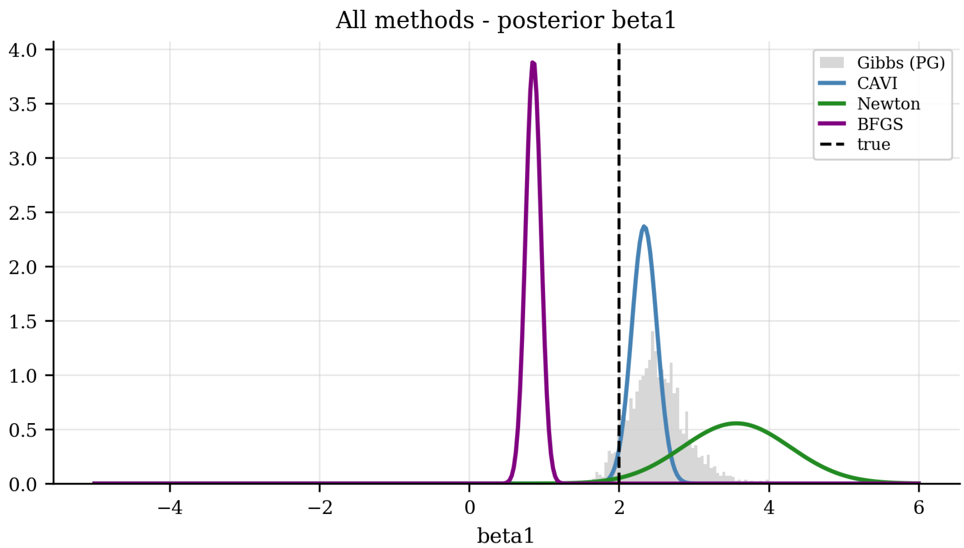

Figure 18: All methods — posterior distribution for β_1.

5.12 Timing Comparison

Figure 19 shows the computational cost of each method on a log scale, with annotations giving the speedup factor relative to the PG Gibbs sampler.

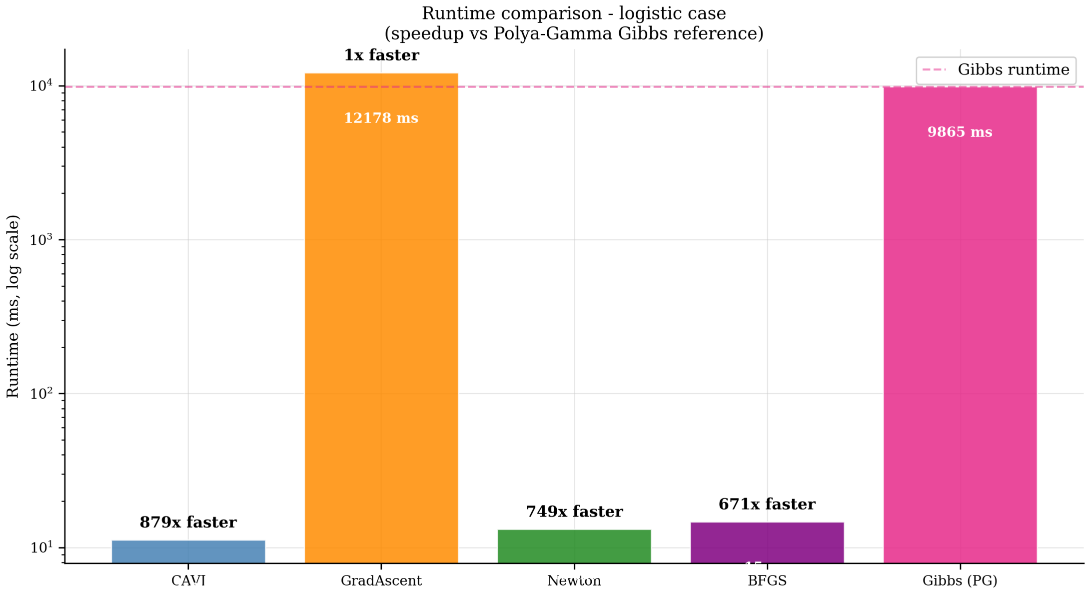

Figure 19: Computational cost on a log scale. Annotations give speedup relative to the PG Gibbs reference (dashed line). Timing comparisons between VI methods and the PG Gibbs sampler are qualitative: the PG sampler requires $n \times T = 10\,000$ Gamma variates per iteration while VI methods use only gradient evaluations.

6 Discussion

The logistic case study is a more demanding setting for mean-field VI than conjugate linear regression. The posterior $p(\boldsymbol{\beta} \mid \boldsymbol{y})$ is not Gaussian, so the variational family $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is an approximation regardless of the quality of the optimum found. There is no mean-field factorisation over separate parameter groups — $\boldsymbol{\theta} = \boldsymbol{\beta}$ only — so that source of additional error is absent. The main approximation error comes from fitting a Gaussian to a distribution that is only approximately Gaussian. In practice, with $n = 200$ observations the posterior is sufficiently concentrated that the Gaussian approximation is accurate, as confirmed by the SD ratios of 0.90–1.10.

6.1 Posterior Variance Collapse

Posterior variance collapse is minimal in this model. All four VI methods achieve standard-deviation ratios $\sigma_{\mathrm{VI}}/\sigma_{\mathrm{Gibbs}}$ between 0.90 and 1.10 for both parameters. The mechanism is the tightness of the Jaakkola–Jordan bound: at the optimal $\xi_i^* = \sqrt{\mathbb{E}_q[\psi_i^2]}$, the bound equals the true log-likelihood expectation, so the ELBO objective penalises over-concentration rather than permitting it. CAVI and all gradient-based methods consequently recover both posterior means and posterior standard deviations accurately relative to the PG Gibbs reference.

6.2 Optimisation Behaviour

CAVI is the most efficient method due to its closed-form updates under the Jaakkola–Jordan bound. Gradient ascent converges more slowly but requires only gradient computations. Newton's method converges in fewer iterations but requires a finite-difference Hessian (25 evaluations per iteration), making each step expensive. BFGS provides a good compromise: quasi-Newton curvature information with only gradient evaluations, and converges rapidly due to the smooth, near-quadratic ELBO landscape.

6.3 Model Characteristics

The Bernoulli likelihood is not conjugate to the Gaussian prior, so the ELBO has no closed form. The Jaakkola–Jordan bound resolves this by introducing local variational parameters ξ_i that render the ELBO analytically tractable. The reference distribution is a Pólya–Gamma Gibbs sampler: exact (up to PG truncation) but expensive, requiring $n \times T = 10\,000$ Gamma draws per iteration. All timing figures are from an Alienware 17 R5 (Intel i7-8750H).

For the broader study, this case study is the non-conjugate setting in which conjugacy is unavailable and gradient-based ELBO optimisation becomes important. When conjugacy is available, CAVI should be treated as the natural method and the generic gradient methods should not be expected to improve on it. Here conjugacy is not available, so CAVI and gradient-based methods rely on the Jaakkola–Jordan bound for tractability; the bound provides analytical gradients that make this feasible. Without it, finite-difference gradients at $n = 200$ observations would not scale well.

Conclusion

This report has applied gradient-based optimisation methods to variational inference in Bayesian logistic regression using the Jaakkola–Jordan ELBO. The formulation is an ELBO maximisation problem, but the numerical question is broader than maximisation alone: all four VI methods (CAVI, gradient ascent, Newton, BFGS) successfully maximise the ELBO and recover posterior distributions that closely match the PG Gibbs reference. CAVI is the preferred method; BFGS is the recommended gradient-based alternative when closed-form updates are unavailable.

The main issue is whether a converged variational approximation also gives a credible account of posterior uncertainty. Cholesky reparameterisation keeps the covariance parameters valid during optimisation, while the PG Gibbs sampler provides the reference for posterior spread.

The key conclusion is that optimisation performance and uncertainty quality must be read separately. A method can make progress on the ELBO while still producing a posterior approximation that is too narrow. That distinction is the main lesson of the logistic case study and the reason posterior standard-deviation ratios are reported alongside convergence results.

References

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Polson, N. G., Scott, J. G., & Windle, J. (2013). Bayesian inference for logistic models using Pólya–Gamma latent variables. *Journal of the American Statistical Association*, *108*(504), 1339–1349. https://doi.org/10.1080/01621459.2013.829001

Appendix A VB Posterior Variance Collapse

Why Variance Collapse Happens

Variational inference finds $q(\boldsymbol{\beta})$ by minimising the KL divergence from q to the true posterior $p(\boldsymbol{\beta} \mid \boldsymbol{y})$:

$$\mathrm{KL}[q \| p] = \int q(\boldsymbol{\beta}) \log \frac{q(\boldsymbol{\beta})}{p(\boldsymbol{\beta} \mid \boldsymbol{y})} \, d\boldsymbol{\beta}. \tag{15}$$

This is the forward, or exclusive, KL divergence. It penalises q for placing mass where the true posterior is small, but not for missing posterior tails. The approximation therefore tends to stay near the high-density centre of the posterior and can be too narrow.
In this model the variational family $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is a joint Gaussian over the full coefficient vector — there is no mean-field factorisation across separate parameter groups (unlike the linear case where $q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta})q(\tau_e)$ imposes an additional independence restriction). The only source of collapse here is the KL asymmetry itself.

Why Collapse is Minimal in This Model

The Jaakkola–Jordan bound is tight at the CAVI fixed point: when $\xi_i^* = \sqrt{\mathbb{E}_q[\psi_i^2]}$, the bound equals the true log-likelihood expectation. This means the ELBO objective at the optimum correctly reflects the true likelihood contribution, so the entropy term in the ELBO actively penalises over-concentration. The KL asymmetry is therefore partially counteracted by the bound structure itself.

Observation

All four VI methods achieve posterior standard-deviation ratios $\sigma_{\mathrm{VI}}/\sigma_{\mathrm{Gibbs}}$ between 0.90 and 1.10 for both parameters, confirming that collapse is minimal in this two-parameter logistic model with $n = 200$ observations. No corrective measures (entropy regularisation, variance lower bounds) are required.

Appendix B Cholesky Reparameterisation: Structural Constraint Satisfaction

Why Reparameterisation?

The gradient-based optimisers operate on an unconstrained Euclidean parameter vector, but the variational covariance $\boldsymbol{\Sigma}$ must remain positive definite at every iteration. If the optimiser updates $\boldsymbol{\Sigma}$ directly, a gradient step $\boldsymbol{\Sigma}^{(t+1)} = \boldsymbol{\Sigma}^{(t)} + \alpha_t \nabla_{\boldsymbol{\Sigma}} \mathcal{L}$ can produce a matrix that is not positive definite, causing numerical failure. Reparameterisation encodes the constraint structurally so it cannot be violated.

Cholesky Decomposition

The covariance is written as $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$ where $\boldsymbol{L}$ is lower triangular with strictly positive diagonal entries. For $p = 2$ the factor is

$$\boldsymbol{L} = \begin{pmatrix} e^{\tilde{\ell}_{11}} & 0 \\ \ell_{21} & e^{\tilde{\ell}_{22}} \end{pmatrix}, \tag{16}$$

giving the unconstrained optimisation vector $\boldsymbol{\theta} = (\mu_0, \mu_1, \tilde{\ell}_{11}, \ell_{21}, \tilde{\ell}_{22}) \in \mathbb{R}^5$.
Any $\tilde{\ell}_{11}, \tilde{\ell}_{22} \in \mathbb{R}$ gives $L_{ii} = e^{\tilde{\ell}_{ii}} > 0$, so the resulting $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$ is always symmetric positive definite. No projection, penalty, or clipping is required: any point in $\mathbb{R}^5$ maps to a valid variational Gaussian, and standard optimisation machinery applies without modification.

Appendix C Code Structure

The implementation is in `python/logistic_run.ipynb` (35 cells).

Stage 1 (cells 2–4): Data generation. Saves `results/data/logistic_data.parquet`.

Stage 2 (cells 5–9): ELBO evaluation and gradient verification. Functions `elbo()`, `elbo_grad()`, `unpack()`, `pack()`.

Stage 3 (cells 10–18): Optimisation methods. Functions `run_cavi()`, `run_gradient_ascent()`, `run_newton()`, BFGS via `scipy.optimize.minimize`. Saves `results/data/logistic_opt.parquet`.

Stage 4 (cells 19–24): PG Gibbs sampler with ESS diagnostics. Functions `sample_pg_one()`, `sample_pg_vec()`, `acf()`, `compute_ess()`. Saves `results/data/logistic_gibbs.parquet`.

Stage 5 (cells 25–34): Diagnostics and comparison. Saves `results/tables/logistic_diagnostics.csv` and `.parquet`.

All figures are saved to `figs/` with the `logistic_` prefix to avoid collisions with figures from other notebooks in this project.